

Operator & external-agent entry surfaces — all pass the same binding model

Web console

TUI workbench

MCP server · 25 tools

A2A endpoint

Plugin host (hot-reload)

Agent half

Harness registry — 14 engines
availability allow-list probe

proposals

Teams & channels

lead supervision · task tree · watchdog

Memory stack

FTS5+vector · RRF/MMR · compaction

Governed bridge

Agent tool surface — 48 tools
semantic cards ground physical units

Node bindings

manual / auto / unbound gate the path

Governed write path (Alg. 1)

approval → guardrail → interlock →
readback-verified write → journal

Backstop (30 s cadence, breach ≥ 3):
bounded rollback $K=2 \rightarrow$ human

journal · state

verified write $\leftrightarrow$ readback

Industrial half

Protocol drivers

Modbus TCP/RTU · OPC UA · MQTT · HTTP

Edge runtimes — 3 decoupled intervals

per-node schedule · in-flight exclusion

Gateway sweep

rotating cursor · starvation-free quota 64

Alarm chain

hysteresis · debounce · recipe-window overlay

Line / recipe management

versioning · known-good · active runs

film-line
scenario

cast-film physics
twin + ODE

WS push

scene-edit writeback

Audit plane

signed journal · ops log · attribution

Storage — one data domain

TimescaleDB · MinIO · SQLite (WAL)

Device-twin registry

modelRef · boundAgentId · 3 state planes

3D scene (three.js)

live render · twin observes

Data, audit & digital-twin plane — one data domain